

Heart Disease Risk Analysis

Project Summary, Methodology & Tech Stack Reference Guide

1. Project Overview

This is an end-to-end data science and machine learning project exploring the interplay between clinical lab markers (blood pressure, cholesterol, HbA1c, ST depression) and lifestyle factors (sleep duration, stress levels, physical activity, diet, and wearable ownership) in predicting heart disease risk.

Core Research Question: *Do everyday lifestyle metrics add genuine, independent predictive power beyond traditional clinical diagnostics?*

2. Dataset Profile

File: heart_disease_risk_2026.csv | **Volume:** 9,001 patient records, 26 features | **Target:** has_heart_disease (Binary: 0 or 1, ~70/30 class distribution).

Categories: Demographics, Blood Pressure, Lipid Profile, Blood Glucose, Vitals, Symptoms, Lifestyle & Wearables.

3. Methodological Workflow

- **Exploratory Data Analysis (EDA):** Statistical distribution checks, correlation analysis, and targeted hypothesis testing. Partial correlation residualisation was performed to evaluate whether stress and sleep remain statistically significant after controlling for BP and lipids.
- **Hypothesis Testing:** Non-parametric Mann-Whitney U tests and rank-biserial correlations assessed differences between wearable owners and non-owners.
- **Preprocessing:** Categorical one-hot encoding, boolean conversion, and a rigorous 60 / 20 / 20 stratified train/validation/test split. Feature standardisation (StandardScaler) applied specifically for distance/linear models.
- **Predictive Modeling:** Trained Logistic Regression (baseline), Random Forest (ensemble), and XGBoost (gradient boosting). Primary evaluation via ROC-AUC to prevent accuracy paradox on imbalanced classes.
- **SHAP & Feature Ablation:** TreeExplainer computed exact Shapley contributions. Controlled ablation experiments compared: (1) Lifestyle Only, (2) Clinical Only, and (3) Full Feature Set.

4. Key Findings

1. **Clinical labs dominate baseline risk:** Markers like ST depression, HbA1c, and age account for the largest proportion of predictive importance.
2. **Lifestyle adds measurable incremental signal:** Combining lifestyle features with clinical labs improved test ROC-AUC, confirming lifestyle data provides complementary risk insight.
3. **Independence validated:** Stress scores and sleep deficits showed statistically significant partial correlation with heart disease independent of cholesterol and blood pressure.
4. **Wearable caveat:** Merely owning a fitness tracker showed negligible effect size regarding actual activity or risk reduction.

5. Technologies & Python Libraries Used

Library / Tool	Role & Description
Python	Core programming language used for the entire data pipeline, statistical computation, and modeling.
pandas	Data manipulation and analysis: reading the CSV, handling datatypes, grouping, and aggregations.
numpy	Vectorized numerical computations, linear regression matrix math for residualisation, and array operations.
scikit-learn	Core ML toolkit: train/val/test splitting, StandardScaler, Logistic Regression, Random Forest, ROC-AUC, precision, recall, confusion matrix.
XGBoost	Optimized distributed gradient boosting framework used as the primary high-performance classifier.
SHAP	Game-theoretic Shapley Additive Explanations used for local and global model explainability (summary beeswarm, dependence plots).
scipy	Scientific computing: Mann-Whitney U test (non-parametric comparison) and Pearson correlation coefficients.
matplotlib	Foundational data visualization library for rendering figures, axes, bar charts, and ROC curves.
seaborn	High-level statistical visualization: custom risk heatmaps, probability density distributions, and palette styling.
Jupyter / ipykernel	Interactive computational notebook environment enabling reproducible execution and visual report rendering.
uv	High-performance Python package manager used for deterministic environment isolation and dependency resolution.

6. Governance & Limitations

Synthetic Dataset: Procedurally simulated cohort created for machine learning exploration. It does not represent clinical patient trials and must not be utilized for real-world medical diagnosis or treatment decisions.

Cross-Sectional Nature: Lacks longitudinal time-series data on disease progression over decades.

Model Calibration: Optimized for discrimination (ranking via ROC-AUC) rather than absolute calibrated clinical probabilities.